

KRISHNA SANJAY V ADDI

MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Machine Learning Engineer with an M.S. in Data Science and Analytics (GPA: 3.89). Builds production-grade computer-vision, NLP, RAG, and LLM systems from training and evaluation through APIs, PostgreSQL backends, Docker, CI/CD, monitoring, and edge deployment. Hands-on experience includes YOLOv11, ByteTrack, Vision Transformers, TensorRT INT8 quantization, FastAPI, PyTorch, and automated test pipelines.

TECHNICAL SKILLS

ML Engineering	Python, PyTorch, Scikit-Learn, Pandas, NumPy, OpenCV, CUDA/TensorRT, model evaluation
Computer Vision & AI	YOLOv11, ByteTrack, OCR, Vision Transformers, LLMs, RAG, LangChain, NLP, multi-agent systems
Production & MLOps	Docker, CI/CD, pytest, model monitoring, experiment tracking, Roboflow, AWS SageMaker, Git
Backend & Data	FastAPI, Node.js, React/Next.js, SQL, PostgreSQL, Supabase, ETL, Plotly, Power BI

EXPERIENCE

POWO Studios

Boca Raton, FL

ML Engineer

Feb 2026 - Present

- Owned the inventory MVP architecture, research, and end-to-end implementation in a two-person team, advancing the system to a mid-September 2026 pilot.
- Engineered the YOLOv11s, OCR, ByteTrack, zone-logic, and Supabase/PostgreSQL pipeline; achieved 91.6% mAP@50 and routed uncertain detections to a vision-language fallback for reliable event capture.
- Automated explainable inventory alerts and vendor reorders through an LLM service, targeting 40% less manual inventory effort and faster stockout response.
- Built AthenaText's FastAPI/PostgreSQL backend and React/Next.js developer console, giving developers self-service messaging access across 24 console pages and seven SDK languages.
- Hardened delivery with Docker, CI/CD, scheduled jobs, drift monitoring, and automated regression coverage across 498 API and console test files, reducing release risk.

RedFiber

Navi Mumbai, India

Machine Learning Intern

Apr 2023 - Jul 2023

- Co-developed the Aizer Automation System with foundation models and NLP, reducing manual research time by 40%.
- Built SpaCy, NLTK, and Regex preprocessing plus FAISS retrieval modules, converting unstructured enterprise content into production-ready semantic search.
- Created ROUGE, BLEU, and semantic-similarity evaluation benchmarks, establishing repeatable quality gates for generated outputs.

SELECTED PROJECTS

YOLOv11s INT8 Quantization for Jetson Orin NX

- Converted YOLOv11s from FP32 through FP16 to INT8 TensorRT engines for low-latency edge inference on NVIDIA Jetson Orin NX.
- Engineered CUDA/TensorRT fused plugins and a sensitivity-driven selective FP16 fallback for precision-critical layers.
- Built a per-class mAP validation suite with calibration sweeps and a 56-test host/device split script.

KardioMedicus - Clinical Decision Support System (ECG)

- Fine-tuned a Vision Transformer on ECG spectrograms to 89% classification accuracy and integrated guideline-grounded RAG with SHAP heatmaps.

AI Content Engine - Multi-Agent Workflow Pipeline

- Orchestrated specialized Claude 3 agents for briefs, copywriting, SEO, and A/B variants; built deterministic ranking and offline mock agents for zero-cost CI testing.

TaxiBot - Natural-Language BI Copilot

- Built a Llama 3 Text-to-SQL assistant for 50K+ NYC taxi-trip records with conversation state, prompt validation, executable SQL generation, and Plotly chart dispatch.

PUBLICATIONS

- Vaddi, K. S. (2025). Revolutionizing Diabetic Retinopathy Detection. CRC Press, 1st Edition.
- Vaddi, K. S. (March 2024). Enhancements in the World of Digital Forensics. International Journal of AI.

EDUCATION

Florida Atlantic University

M.S. in Data Science and Analytics (GPA: 3.89)

Boca Raton, FL

Aug 2024 - May 2026

Bharati Vidyapeeth

B.Tech in Computer Science and Business Systems (GPA: 4.0)

Navi Mumbai, India

2020 - 2024